A small JEPA world model on a low-cost arm: the encoder reads the outcome, the predictor adds nothing to it

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Abstract

We trained a small JEPA world model in the LeWorldModel recipe for $3.30 of rented GPU time. The training data are teleoperated demonstrations of one tabletop task on an SO-101 arm. We then re-tested on this model the uses of such models that recent papers propose. Each use has a threshold in a public register. We tested the encoder on policy rollouts the model never saw. A linear probe on its last-frame latent separates success from failure about as well as a hand-built colour judge. It does better than a frozen DINOv2 encoder and better than the commanded joints. The predictor adds nothing to this. At the grasp, the imagined latent reads the outcome at 0.70 and the current latent at 0.82. Every use that needs the predictor to roll forward misses its threshold. It still does when we train the same recipe on 45 to 2,909 episodes, which adds a second dataset and then another lab's rollouts. The controls are exploratory and come from one rig and one model per size.

1 Introduction

LeWorldModel (LeWM) [1] trains a world model in two parts. The encoder maps camera frames to a latent vector. The predictor takes the current latent and a sequence of actions and imagines the latents that follow. Both are trained together from pixels with a joint-embedding predictive (JEPA) loss. Most of the uses that later papers propose need the predictor. LeWM itself ranks a policy's candidate action chunks by their imagined outcome. ContactGuard [2] reads the outcome of a grasp before it happens, HWM [3] and H-JEPA [4] plan over several time scales, and other papers flag a failure when the observed latent drifts from the predicted one. Each of these papers reports that its use worked on its own robot and data.

We had one such model, trained on our own arm. We ran every use we could find on it. Each re-test is a row of a register with a threshold, a dataset, a result and a verdict. Where we managed it, the threshold and the decision rule went into the repository before the result existed. That commit order is what we mean by pre-registration. All 20 rows match a pre-registration record in the commit history. In 19 of them the bar was in place before the result. The two-level hierarchical progress use had it in the same commit. One more row is not a use of our model. It times the pretrained RoboJEPA [5] and V-JEPA 2.1 [6] encoders on our on-board computer. 14 of the 20 rows fail or are no-go. Table 1 lists all of them.

The first version of this paper argued something completely different. We read the failures as a sign that the latent mostly encodes the arm's pose. Then a frozen image encoder or the commanded joints would do at least as well. We ran these controls on 69 policy rollouts. The encoder read how the episode went better than either control. We modified the claim around the predictor.

We come back several times to one moment, the grasp anchor. It is 15 frames before the gripper first closes. There the current latent reads the outcome at 0.82 and the imagined latent at 0.70.

We added the controls after the data. Each readout is a linear probe with leave-one-session-out (LOSO) cross-validation on the 69 rollouts (44 success, 25 failure, 13 sessions). The author gave the labels at rest. The model was trained on teleoperation only and never saw these rollouts. No threshold was committed for these comparisons, so everything this paper says about the split between encoder and predictor is exploratory. 3 of the 3 register passes have their bar committed before the result. The register still marks the outcome-readout verdict as retrospective.

2 Setup

The robot is one SO-101 arm [7] driven through LeRobot [8], with a top and a wrist camera. The task is to put a bowl into another bowl. We trained our model on `ravediamond/so101_bowls_all_trimmed`, split by grid layout into 157 training and 25 validation episodes. The evaluation rollouts come from a separate dataset, `ravediamond/so101-bowls-policy-rollouts`. They are

*Code, data pointers and every number's source: https://github.com/ravediamond/paper-jepa-world-model.

SmolVLA [9] runs recorded after our model was trained.

Our model follows the LeWM recipe [1], with a multi-view ViT encoder, an action-conditioned predictor and the SIGReg regulariser. We compare it with four cheaper signals. The frozen DINOv2-S control is a frozen DINOv2 small encoder [10], mean-pooled over the top and wrist views as in DINO-WM [11]. The joints control is the commanded joint vector. The colour judge is a hand-built rule on the final frame with no learning. Policy identity is the checkpoint that ran the episode.

In October we found a loading bug. A timing test on the on-board computer loaded our checkpoint with `strict=False`. The image had transformers 5.5.4, and under that version 192 encoder tensors did not match. They were dropped without an error, so the test timed a random encoder, and we voided its latents. We then traced how every other use loaded the model. 26 ran the trained model and 1 ran a random one, the timing test. The world-model block of VLA-JEPA, which one diagnostic below uses, loads all 351 of its keys.

3 The register

Table 1 lists the 20 uses. The takes column says what each one needs from our model: the encoder alone (7), the predictor (8) or both (5).

Among the encoder uses, reading the outcome works. The outcome readout at rest passes. At the grasp anchor the current latent reads the grasp at 0.82. Reading progress does not work. Four uses ask the latent for progress or task stage, and all four fail. They are progress monotonicity, two-level hierarchical progress [4], the latent stage tracker and SURVEYOR's latent-distance scale [14]. We also ran the pose probes of LeJEPA [13]. A linear readout recovers about half of the joint variance from our encoder (R^2 0.565). It recovers more from the frozen DINOv2-S control (0.867).

The predictor does predict its own latent. At K = 30 its open-loop error is 0.31. Copying the last latent gives 0.89. The uses of this prediction fail. In chunk ranking [1] the model puts the executed chunk first in 0.60 of anchors. The threshold was 0.8. The commanded actions alone reach 0.68–0.90. The surprise monitor catches 2/8 at 2/42 failures. The hierarchical planner [3], the conformal surprise band, re-ranking, the risk head [12] and the calibrated reflex also miss their thresholds. Table 1 gives the margins.

4 What the encoder knows

We tested whether the encoder carries anything that a frozen encoder or the joints do not. Table 2 sets our model beside the four other signals on the same 69 rollouts. Every probe is the same leave-one-session-out logistic regression, fit against the author's labels (`repro/gaps/scripts/analyze_controls.py`):

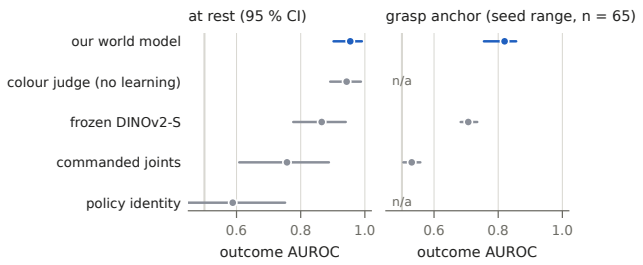


Figure 1: Table 2 as a dot plot. Left: at rest, with the 95 % CI. Right: at the grasp anchor, with the seed range (not a CI).

```
def loso(X, y, groups):
    p = np.zeros(len(y))
    for g in np.unique(groups):
        te = groups == g
        sc = StandardScaler().fit(X[~te])
        m = LogisticRegression(C=1.0, class_weight="balanced",
     max_iter=5000).fit(sc.transform(X[~te]), y[~te])
        p[te] = m.predict_proba(sc.transform(X[te]))[:, 1]
    return p
```

At rest the interval is a 95 % episode-bootstrap CI. At the anchor it is a bootstrap of the mean over 5 seeds of the out-of-fold AUROC, with episodes resampled jointly across the seeds (2,000 resamples, percentile).

At rest our model is level with the colour judge. The two CIs overlap. Our model is above the frozen encoder and the joints. At the grasp anchor no judge applies, and our model is above both controls. The pose probes hold the joints better than our encoder does. They hold the outcome worse. On these rollouts the encoder has outcome information that the pose alone does not give.

The anchor column depends on how the five seeds are combined. If we average the predictions before scoring, our model gets 0.8377, frozen DINOv2-S 0.7175 and the joints 0.5346. With seed 0 alone they get 0.7554, 0.7024 and 0.5574. The gap to the frozen encoder is smaller with seed 0. The order is the same.

On SURVEYOR's scale check the controls do better than our model. Frozen DINOv2-S has τ 0.3601 and the joints τ 0.0265, and both pass. Our model has τ 1.2488 and fails. SURVEYOR reports 0.20. The task and the rollouts are the same for all three features. We take the failure as a property of our latent space.

The register's own outcome-readout row reports 1.000. That number was scored against the colour judge's labels on 87 rollouts. Scored against the author's labels on the 69 rollouts, the same readout gives 0.9545.

5 Where the predictor fails

ContactGuard [2] finds that probes on the imagined latent beat probes on the current latent for grasp tasks. We expected the same at our grasp anchor, and the pre-registered threshold asked the imagined latent for ≥ 0.85, $\geq +0.05$ over the current one. We found the reverse. With

Table 1: The register: every use of our world model re-tested here (with the frozen-DINOv2 and RoboJEPA rows beside it), what it takes from the model, whether its bar came before the result, and the verdict. Read the takes column against the verdict: the predictor uses fail. Bar, result and verdict are quoted from repro/register/expected/FEASIBILITY.md §2.

use	re-tests	takes	bar first	bar	result	verdict	code
open-loop latent prediction	[1]	predictor	yes	beats copy-last; ratios ≥ 1.3	open loop K = 30: 0.31 vs copy-last 0.89; timing shifts 1.99 / 1.58; rank 24–30	**pass**	G0
imagined- vs current-latent outcome probe (first run)	[2]	both	yes	imagined beats current	imagined 0.72, current 0.68; policy identity 0.76	not proven (policy confound)	C1
action-chunk ranking	[1]	predictor	yes	top-1 ≥ 0.8	top-1 best 0.60; action-only 0.68–0.90, regret ≈ 0 (52)	**fail**	G1b
arm-pose linear probe	[1, 11]	encoder	yes	> 0.57	arm-pose R^2 ours 0.57, **DINOv2-S 0.92**	DINOv2-S pass	G1
chunk ranking with a frozen-DINOv2 world model	[11]	predictor	yes	≥ 0.8	DINOv2-S top-1 0.55–0.61	**fail**	G1-W2
progress monotonicity and done detection	this project	encoder	yes	$\rho \leq -0.8$; AUROC ≥ 0.95	monotonicity -0.32 to -0.50; done AUROC ours 0.97 (easy contrast); 32 % false stalls	T1 **fail**, T2 sanity	G6a
outcome readout at rest; imagined-latent anticipation	[2]	both	yes	≥ 0.90; $+0.10$	readout at rest AUROC ours 1.000, DINOv2-S 0.976; anticipation $+0.179$ over policy identity	**pass** (retrospective)	G10
two-level hierarchical progress	[4]	encoder	same commit	≤ -0.80; ≤ 10 %	H-JEPA level 2 ρ -0.727; false stalls 52 %	**fail**	G6b
surprise monitor	this project	predictor	yes	$\geq 5/8$ at $\leq 1/20$	L2 surprise 2/8 at 2/42 (AUROC 0.86); drift 0/8; live reflex 42/42 successes flagged	**fail**	W1-A
latent stage tracker	this project	encoder	yes	≤ 2 %, ≤ 10 %	22.1 % premature, 31.1 % missed	**fail**	W1-B
calibrated surprise reflex	this project	predictor	yes	recall ≥ 50 %	3/25 failures at 1/44	**fail**	G11-B-cal
imagined-latent grasp probe	[2]	both	yes	≥ 0.85, $\geq +0.05$	imagined 0.70 < current 0.82	**fail**	W1-C1
conformal surprise band	this project	predictor	yes	$\leq 1/20$, ≥ 50 %	successes 22.7 % flagged, failures 46 %	**fail**	W1-C2
re-ranking chunk candidates	[1]	predictor	yes	≤ 0.10	regret 0.2631 vs ours 0.2630, action-only -0.0075	**fail**	W1-R
action-conditioned risk head	[12]	both	yes	$\geq 5/8$ at $\leq 1/20$	first version 0.93 withdrawn (time leak); second version 3/8 at 5/42, AUROC 0.79 vs shuffled 0.71	**fail**	REAP
inverse dynamics from a latent pair	this project	encoder	yes	random-init < half ours	latent pair $+0.748$ over one frame, but random-init ours 1.254 vs ours 1.180; beyond state $+0.007$	**not proven**	§30
linear identifiability of the latent	[13]	encoder	yes	$R^2 < 0.80$; ≥ 0.20	ours mean R^2 0.565, anisotropy 0.639; DINOv2-S 0.867 / 0.301	B1–B3 met	LeJEPA
latent-distance scale for plan consumption	[14]	encoder	yes	$\tau \leq 0.40$; $\geq 5/8$ at $\leq 1/20$	τ 1.249 (paper 0.20); monitor 3/8 at 6/42	**fail**	SURVEYOR
hierarchical vs flat prediction horizon	[3]	predictor	yes	crossover in [1, 2] s	no crossover: high level 0.76 vs ours 0.56 at 2 s	H1, H2 **fail**	HWM
on-device latency of RoboJEPA / V-JEPA 2.1	[5, 6]	both	yes	≤ 200 ms; ≥ 15 %	encoder 526 ms (2 views); $\leq$ copy-last + 12 %	**no-go**	RoboJEPA

Table 2: Outcome AUROC of each signal on the same 69 rollouts (exploratory: controls chosen after the data). No judge applies at the anchor.

feature	at rest	grasp anchor (65 scored, 21 fail)
our world model	0.9545 [0.9023, 0.9908]	0.8199 [0.7141, 0.9041]
colour judge (no learning)	0.9432 [0.8919, 0.9878]	—
frozen DINOv2-S	0.8655 [0.7767, 0.9405]	0.7067 [0.5816, 0.8269]
commanded joints	0.7573 [0.6084, 0.8880]	0.5305 [0.3709, 0.6855]
policy identity	0.5882 [0.4084, 0.7515]	—

the same encoder and the same anchor, the current latent reads the outcome at 0.82 and the imagined latent at 0.70. The row fails.

We also ran the distillation probe on our model's inverse-dynamics term, pooled over the 25 teleoperation validation episodes. The trained model scores 0.7476. Its random initialisation scores 0.7974. The difference is -0.0499, with a 95 % paired episode-bootstrap CI of $[-0.7403, +0.6559]$ (`gap5_pooled_rerun.json`). The term depends on the machine. The same untrained net-

Table 3: VLA-JEPA world-model loss at 5 anchors with true, shuffled (Δ shuffled) and zeroed action tokens (exploratory spot check). The token change is the relative L2 of the shuffled tokens.

anchor (episode / frame)	true loss	Δ shuffled (%)	token change (%)	Δ zeroed (%)
8 / 402	1.3031	+0.014	21.1	+2.22
10 / 227	1.4641	+0.088	27.6	+0.50
23 / 327	1.4215	+0.036	21.1	+0.87
59 / 302	1.4637	+0.030	25.5	+1.28
65 / 302	1.5236	−0.070	22.0	−0.38

work scored 1.4383 on one machine and 0.7974 and 0.8042 on two others. It is sensitive to the BLAS library and to ridge conditioning.

The surprise monitor has a failure-class AUROC of 0.775, below its sanity bar. We checked it for a loading or scoring bug. The same scorer reproduces the open-loop corruption ratios at K = 30 (`M1_diag_g0_repro.json`). Cross-episode gives 3.428, against 3.341 in the original run. Constant-first gives 2.558, against 2.543. We found no bug.

We ran an action-shuffle check on the predictor of VLA-JEPA at 5 anchors (Table 3, from `M3_diag_action_shuffle.json`). Action tokens from another episode differ from the true ones by 21–28 % in relative L2. They move the loss by about 0.05 %. Zeroing the tokens moves it by about 1.05 % on average. The forward pass is deterministic. The predictor barely uses its actions. Its failure-class AUROC is 0.501, below the same sanity check, and we leave its table unread under the frozen bar.

The outcome-readout gate has one pass that uses the imagined latent. In its anticipation step the imagined latent beat policy identity by +0.179. We set that bar after the result. We then reran the idea twice with the threshold committed first, once in the grasp probe and once in the calibrated surprise reflex, and both runs fail.

6 Does more data change this?

All of this comes from one model trained on 157 episodes. We trained the same recipe at three other sizes, for the same 81,068 gradient steps as the original model, and ran the same probes on each checkpoint (Table 4). The smallest model sees 45 of the 157 episodes. The original model is the second row and was not retrained. The third adds 653 episodes of a second cubes-and-bowls dataset of ours at the same frame rate. The fourth adds the 2,099 policy rollouts of ArmnetBench, which come from another lab with other cameras and which we kept at their native 20 Hz, so the step to the last row changes the data and the setup together. The probes, thresholds and decision rule were committed before any of these runs (`repro/data-scale/PROTOCOL.md`).

No predictor use leaves its floor. The commanded actions rank the executed chunk first at 0.873 at d = 30, and the model stays below them at every scale with a CI that excludes zero. Its best cell goes from 0.422 at 45 episodes to 0.605 at 157 and then down to 0.279 and 0.230 when the other datasets are added, so it never gets near the 0.8 threshold. We did not test why more data made the ranking worse. The added episodes are other tasks and another rig, and the ranking is scored on the same 25 bowls episodes at every scale, so the added data may take capacity from the task we score. The surprise monitor catches 1, 2, 1 and 0 of 8 failures, against the threshold of 5.

The encoder does not move much either. The joints R^2 stays between 0.556 and 0.573. The rest readout stays level with the colour judge (0.943) at every scale: the CI of the paired difference includes zero in each row, and the lowest value, 0.897, belongs to the largest model. The beyond-state probe has the same verdict at every scale, not proven. The latent now and 5 frames later predict the commanded action better than the latent now alone (the gain is at least 0.62). What they add to the measured joints stays between 0.006 and 0.009, against a threshold of 0.02, and the untrained encoder gains 1.25 from the same extra latent, which is not under half of what any trained one gains.

This is one run per size with one seed. The decision rule asks for two seeds at the smallest and the largest size, and for a predictor trained on frozen DINOv2-S features at every size as the floor of the surprise monitor. We ran neither. Read it as four runs in which no use of the predictor improves with data, not as a statement about the model class.

7 Related work

LeWorldModel [1] trains the recipe we use end to end from pixels and plans with it in Push-T and in simulation. We tested its two planning uses on our arm, and both fail. LeJEPA's identifiability analysis [13] predicts that SIGReg latents identify the state up to anisotropy. Our pose probes support this in part. DINO-WM [11] plans on frozen DINOv2 [10] patches, and our frozen control uses the same design. We timed V-JEPA 2.1 [6] and RoboJEPA [5] on our on-board computer, and both are too slow. ContactGuard [2] finds imagined-latent probes better than current-latent ones on grasp tasks. On our arm we find the reverse. We also re-tested ReAP's action-conditioned risk [12], the hierarchical planners of HWM [3] and H-JEPA [4], and SURVEYOR's latent-distance scale [14]. All of them miss here, at the original data scale. SaiVLA-JEPA [15] reports that a training-only future-latent head adds +0.03 points to control on 10,030 paired episodes. HIJP [16] finds LeWM below goal-conditioned behaviour cloning in simulation.

Table 4: The probe battery on one model per training-set size. Chunk ranking is the top-1 of 8 candidates at the longest horizon (d = 30) with its gap to the commanded-actions ranker (0.873 top-1 on the same anchors) and the 95 % episode-bootstrap CI of the paired gap. The beyond-state columns come from a ridge probe that predicts the commanded action from the latent now and 5 frames later: the first number is its residual R^2 above the latent now alone or paired with another episode's latent, the second is what the two latents add to the measured joints. The surprise monitor column is the failures caught of 8 at 2 of 42 false interventions. The rest readout is scored on the 69 rollouts against the author's labels.

training episodes (frames)	joints R^2	chunk ranking, d = 30	beyond-state gain / over joints	surprise monitor	rest readout AUROC [CI]
45 (15,032)	0.556	0.403, gap -0.471 [-0.539, -0.405]	1.33 / 0.006	1 of 8	0.931 [0.862, 0.982]
157 (53,015)	0.565	0.605, gap -0.269 [-0.334, -0.204]	0.75 / 0.007	2 of 8	0.955 [0.902, 0.991]
810 (275,015)	0.565	0.273, gap -0.600 [-0.652, -0.545]	0.63 / 0.007	1 of 8	0.956 [0.899, 0.996]
2,909 (1,282,161)	0.573	0.222, gap -0.651 [-0.696, -0.602]	1.17 / 0.009	0 of 8	0.897 [0.813, 0.964]

8 Limitations

We chose the probes for the encoder and predictor split after the data, so the split is exploratory. They are fitted on 69 rollouts that one operator, the author, labelled. The model never saw these rollouts, and every probe is cross-validated by session. To confirm or reject the split we need a fresh set of rollouts scored under a threshold committed first. That takes a day on the arm. The register results come from one model at one data scale. The data-scale runs add three sizes, one run and one seed each. A second seed at the smallest and the largest size, the frozen-DINOv2-S and random-encoder predictors at each size, the 90-episode size and a successes-only variant of the largest were not run. Three controls were not run. The surprise monitor and the imagined arm of the grasp probe would each need a predictor trained on frozen DINOv2-S features or on the joints. Progress monotonicity with the controls would need a reimplementation. In the register the outcome-readout verdict is retrospective, and the two-level hierarchical progress use has its bar in the same commit as its result. Ordered by commit alone, without the audit, the register reads 13 bar-first and 6 same-commit rows.

9 Conclusion

We tested 20 published uses on a $3.30 world model. At its data scale the encoder reads the outcome and the predictor adds nothing to that reading, and training on up to 2,909 episodes did not change either. In chunk ranking the commanded actions did better than the predictor, so they are worth keeping as a baseline for any model of this size.

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